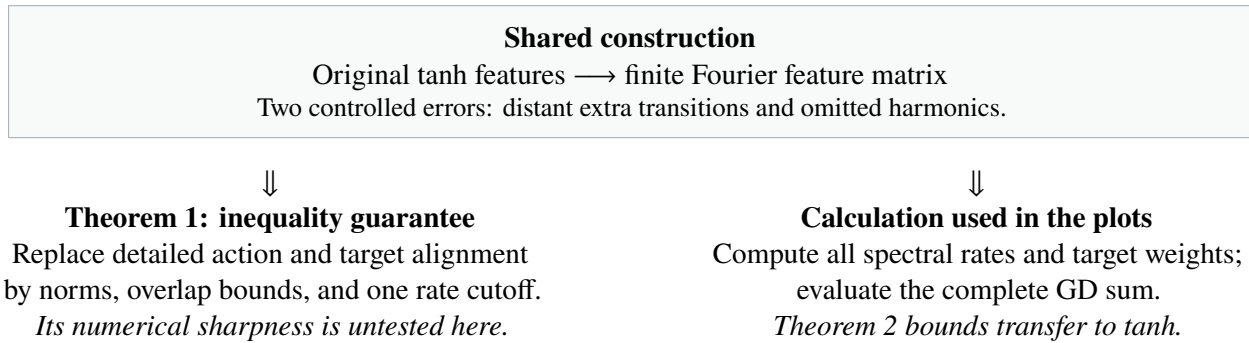


From gamma to slow target directions: the complete argument

Small gamma smooths the features. Whether that slows readout learning depends on the fixed dictionary geometry and on which directions the target needs. This note proves an explicit gamma-dependent lower bound and explains a separate, more detailed calculation that accurately predicts the measured training times.

We computed the full target-weighted spectrum of the Fourier-constructed kernel for every plotted gamma. Here “target-weighted spectrum” means the kernel eigenvalues together with the target’s squared projections onto their eigenspaces. The predicted curves use all those rates and weights. They do not use Theorem 1’s scalar estimate of slow target energy.

Theorem 1 and the plotted timing predictions are two branches of the argument. Both start from the same Fourier construction. Theorem 1 replaces detailed matrix and spectral information with inequalities to obtain a compact guarantee. The plotted predictions instead compute the singular value decomposition of the constructed feature matrix and evaluate its complete GD curve. Theorem 2 bounds the difference between that curve and the original tanh model. The measured delay factor of 986.59 tests this second branch; the numerical sharpness of Theorem 1 on that experiment has not been established.



Notation. Throughout this note, μ_i denotes a kernel eigenvalue. The symbol $\lambda = \gamma h$ is reserved for dimensionless bandwidth when the centers have spacing h . The physical slope is γ . The step size is η_γ , the largest kernel eigenvalue is L_γ , and $a_i = \eta_\gamma \mu_i$ is a per-update decay rate. Norms are Euclidean for vectors and spectral for matrices. Every additional symbol is defined where it enters. Earlier notes used λ_i for eigenvalues and β for bandwidth; this note supersedes that notation.

Reading map: assumptions, approximations, and calculations

The training problem is fixed before any approximation. Inputs, centers, and one common positive slope specify the frozen tanh features. Only raw readout coefficients and a bias are trained, starting at zero, with squared loss and a fixed GD step satisfying $0 < \eta_\gamma L_\gamma \leq 1$. These are assumptions of the learning-time statements. They do not describe training hidden features or Adam. Equispaced centers are used in the main experiment but are not required by the finite construction. The finite training grid defines the problem; no continuous-data or infinite-width limit is being approximated here.

There are two changes to the feature function (Lemma 2). First, an auxiliary repeating square wave replaces the single step before smoothing; its distant extra transitions introduce an error. Second, we keep only Q of its smoothed Fourier harmonics; the omitted harmonics introduce another error. Their sum bounds the difference from the original tanh feature. Increasing the half-period T controls the distant transitions, while increasing Q at fixed T controls the

omitted harmonics. This is the model approximation shared by both branches. Smoothing the original step into tanh is an exact identity (Lemma 1).

Theorem 1 adds inequality slack (Sections 5–6). It replaces matrix action by separate operator norms and a worst-case multiplier, bounds how much a weak direction can overlap fast modes, bounds the target’s overlap with slow modes, and then treats all slow rates as one cutoff. These are valid inequalities whose gaps need not vanish when the Fourier approximation becomes accurate. They do not arise from failing to compute enough eigenvalues. The theorem deliberately avoids needing the full target-weighted spectrum in its bound.

The plotted forecasts use the other branch (Section 7). We assemble the finite Fourier feature matrix, compute its full singular value decomposition, project the target onto the resulting sample-space modes, and sum every mode’s exact GD decay for that approximate model. This calculation is repeated for each gamma. In exact arithmetic, diagonalization and the GD sum introduce no further model approximation. Theorem 2 adds an error envelope to transfer the result to the original tanh problem. Its envelope can itself be conservative. A full spectrum of the original tanh matrix was also computed as an independent reference.

Arithmetic error is a third, separate issue (Section 9). Computer evaluation of waves, multipliers, products, and singular vectors uses finite precision. The analytic Fourier remainder alone does not enclose that rounding. The primary timing endpoints received an independent interval-arithmetic check. The plotted spectral masses are floating-point diagnostics.

Figure 1 uses the computed spectrum and its one-cutoff corollary; Figure 2 compares the full spectral forecast with training. Neither figure evaluates Theorem 1’s estimate from its target-overlap and geometry constants.

1 The model and the optimization question

Fix training inputs $x_1, \dots, x_m$, centers $c_1, \dots, c_W$, and a nonzero target. Only the raw readout coefficients are trained:

$$f_{\gamma, \theta}(x) = b + \sum_{j=1}^W w_j \tanh(\gamma(x - c_j)), \quad \theta = (b, w_1, \dots, w_W)^T.$$

Normalize both the samples of the features and the target by $\sqrt{m}$:

$$(J_\gamma)_{i0} = m^{-1/2}, \quad (J_\gamma)_{ij} = m^{-1/2} \tanh(\gamma(x_i - c_j)), \quad y_i = m^{-1/2} f^\star(x_i).$$

The loss is $\frac{1}{2} \|J_\gamma \theta - y\|^2$, exactly half empirical mean squared error. Define

$$K_\gamma = J_\gamma J_\gamma^T, \quad L_\gamma = \|K_\gamma\|, \quad \theta_0 = 0.$$

The bias column has norm one, so $L_\gamma \geq 1$. Use a prescribed stable step $0 < \eta_\gamma L_\gamma \leq 1$. Our archived experiments have $\eta_\gamma L_\gamma \simeq 1/2$. Changing gamma holds the target, inputs, centers, raw coefficient metric, and step-size rule fixed.

Representability asks whether some coefficients fit the target. Optimization asks how many updates starting at zero are needed to find sufficiently accurate coefficients. A target can be representable while the relevant kernel eigenvalues are very small.

An exact example with no spectral ambiguity

Take one center at zero, samples $-s, +s$ with $s > 0$, and target values $-1, +1$. Put $d = \tanh(\gamma s)$. Then

$$K_\gamma = \frac{1}{2} \begin{pmatrix} 1 + d^2 & 1 - d^2 \\ 1 - d^2 & 1 + d^2 \end{pmatrix}.$$

The mean direction $(1, 1)/\sqrt{2}$ has eigenvalue 1. The contrast direction $(-1, 1)/\sqrt{2}$, which is exactly the target direction, has eigenvalue d^2 . These statements follow by multiplying the matrix by the two vectors. Thus $L_\gamma = 1$, while the target's normalized eigenvalue is $\tanh^2(\gamma s)$.

Every positive gamma fits exactly with $b = 0, w = 1/d$. Nevertheless, at step $\eta_\gamma = \kappa \in (0, 1)$,

$$E_\gamma(n) = (1 - \kappa \tanh^2(\gamma s))^n, \quad n_\epsilon = \left\lceil \frac{\log(1/\epsilon)}{-\log(1 - \kappa \tanh^2(\gamma s))} \right\rceil.$$

As $\gamma s \rightarrow 0$, $n_\epsilon \sim \log(1/\epsilon)/(\kappa \gamma^2 s^2)$. A cap $\gamma \leq \bar{\gamma}$ gives the exact lower bound obtained by replacing gamma with $\bar{\gamma}$ in the displayed time; it is attained at the cap. With $\kappa = 1/2$ and $\epsilon = 0.01$, the counts are 925 for $\gamma s = 0.1$ and 14 for $\gamma s = 1$. A constant target instead decays as $(1 - \kappa)^n$, independently of gamma. This proves both the possible optimization obstruction and the necessity of a target condition.

2 What determines training: rates weighted by the target

Let $r_n = y - J_\gamma \theta_n$ and $E_\gamma(n) = \|r_n\|/\|y\|$. The acquisition time is

$$n_\epsilon(\gamma) = \inf\{n \in \mathbb{N}_0 : E_\gamma(n) \leq \epsilon\},$$

with value $+\infty$ if the set is empty. A 1% relative residual means a relative squared residual of 10^{-4} .

Proposition 1 (Exact GD evolution). Let u_i be an orthonormal eigenbasis of K_γ , including its nullspace, with eigenvalues $\mu_i \geq 0$. Set

$$a_i = \eta_\gamma \mu_i, \quad p_i = |u_i^T y|^2 / \|y\|^2.$$

Then $\sum_i p_i = 1, 0 \leq a_i \leq 1$, and

$$\boxed{r_n = (I - \eta_\gamma K_\gamma)^n y, \quad E_\gamma(n)^2 = \sum_i p_i (1 - a_i)^{2n}.} \quad (1)$$

Proof. The gradient is $J_\gamma^T (J_\gamma \theta - y)$. Hence $\theta_{n+1} = \theta_n + \eta_\gamma J_\gamma^T r_n$ and $r_{n+1} = (I - \eta_\gamma K_\gamma) r_n$. Since $r_0 = y$, induction proves the first identity. Expand y in the orthonormal eigenbasis and square its norm to obtain the second. The convention at $n = 0$ is that every mode contributes its initial energy, including a mode with $a_i = 1$. Each factor is nonincreasing, so the residual is nonincreasing. $\square$

The limiting squared residual is $E_\infty^2 = \sum_{\mu_i=0} p_i$, which is at most every finite-time squared residual. This is the projection of the target outside the feature span; it does not compare the tanh model with a step-function model.

For $\eta_\gamma = \kappa/L_\gamma$, the rates are $a_i = \kappa \mu_i / L_\gamma$. Overall kernel rescaling cancels. The relevant object is the distribution of target energy across normalized eigenvalues, rather than the condition number alone. Individual eigenvectors need not be tracked across gamma; within repeated eigenspaces their choice is arbitrary, but total target energy in each spectral subspace is well defined.

Define the **slow target mass** at rate cutoff $t \in (0, 1)$ by

$$S_\gamma(t) = \sum_{a_i \leq t} p_i.$$

Corollary 1 (Slow mass implies a delay).

$$E_\gamma(n) \geq \sqrt{S_\gamma(t)} (1 - t)^n. \quad (2)$$

If $\sqrt{S_\gamma(t)} > \epsilon$, then

$$n_\epsilon(\gamma) \geq \left\lceil \frac{\log(\sqrt{S_\gamma(t)}/\epsilon)}{-\log(1 - t)} \right\rceil. \quad (3)$$

Proof. Keep only the nonnegative terms with $a_i \leq t$ in (1), and use $(1 - a_i)^{2n} \geq (1 - t)^{2n}$. Taking square roots gives (2); solving its necessary inequality for a crossing gives (3). $\square$

This corollary is already target aware. We now connect its slow mass to gamma without assuming Fourier waves are finite-kernel eigenvectors.

3 Gamma is an exact smoothing operation on the features

For a bounded function $\varphi : \mathbb{R} \rightarrow \mathbb{R}$, define the unit-mass density and smoothing operation

$$\rho_\gamma(t) = \frac{\gamma}{2} \operatorname{sech}^2(\gamma t), \quad (\mathcal{S}_\gamma \varphi)(x) = \int_{\mathbb{R}} \rho_\gamma(x - t) \varphi(t) dt.$$

Here φ may be a constant or a step feature. The calligraphic $\mathcal{S}_\gamma$ distinguishes this operation from the scalar slow mass $S_\gamma(t)$.

Lemma 1 (Exact smoothing and multiplier). Constants are preserved, and $\mathcal{S}_\gamma \operatorname{sign}(\cdot - c) = \tanh(\gamma(\cdot - c))$. For $k_{\text{step}}(t, s) = 1 + \sum_j \operatorname{sign}(t - c_j) \operatorname{sign}(s - c_j)$,

$$\begin{aligned} k_\gamma(x, x') &= \iint \rho_\gamma(x - t) \rho_\gamma(x' - s) k_{\text{step}}(t, s) dt ds, \\ (K_\gamma)_{i\ell} &= k_\gamma(x_i, x_\ell) / m. \end{aligned} \tag{4}$$

With convention $\widehat{\rho}(\omega) = \int \rho(t) e^{-i\omega t} dt$, the multiplier is

$$\boxed{M_\gamma(\omega) = \frac{z}{\sinh z}, \quad z = \frac{\pi|\omega|}{2\gamma}, \quad M_\gamma(0) = 1.} \tag{5}$$

Proof. The density has cumulative integral $[1 + \tanh(\gamma t)]/2$. Splitting the convolution at c gives $2 \int_{-\infty}^{x-c} \rho_\gamma(u) du - 1 = \tanh(\gamma(x - c))$. Unit mass preserves the bias. Insert the finite sum defining k_{step} to prove (4). Bounded features and an integrable density justify the exchanges; the step functions need not belong to whole-line L^2 .

For the transform, put $\xi = \omega/\gamma > 0$ and integrate $e^{-i\xi z} / \cosh^2 z$ around a rectangle with imaginary heights zero and π . The vertical integrals vanish as its real endpoints tend to infinity. The upper horizontal integral is $-e^{\pi\xi}$ times the lower one. The double pole at $z_0 = i\pi/2$ has residue $i\xi e^{\pi\xi/2}$: near z_0 , $1/\cosh^2 z = -1/(z - z_0)^2 + O(1)$. Consequently,

$$(1 - e^{\pi\xi}) \int_{\mathbb{R}} \frac{e^{-i\xi u}}{\cosh^2 u} du = -2\pi\xi e^{\pi\xi/2},$$

so the integral is $\pi\xi/\sinh(\pi\xi/2)$. Scaling $u = \gamma t$ and dividing by two gives (5). Evenness handles negative frequencies and continuity gives the value at zero. $\square$

For $z > 0$, $(z/\sinh z)' = (\sinh z - z \cosh z)/\sinh^2 z < 0$: the numerator starts at zero and has derivative $-z \sinh z$. Thus smaller gamma gives stronger attenuation at a fixed nonzero frequency. A cap $\gamma \leq \bar{\gamma}$ implies $M_\gamma(\omega) \leq M_{\bar{\gamma}}(\omega)$, and $M_\gamma(\omega) \sim 2ze^{-z}$ at high frequency.

Equation (4) smooths the first and second input arguments separately. It is an identity before sampling, not multiplication of the finite sample matrix by an unspecified discrete convolution matrix. The identity alone does not order the finite kernels or their target-weighted spectra.

4 Keep the finite geometry: an explicit matrix construction

We need a finite representation in which the gamma-dependent factors are known and the discarded part is bounded. This construction approximates the original nonperiodic tanh features; it does not make the training problem periodic.

Let $R_0 = \max_{i,j} |x_i - c_j|$, choose a half-period $T > R_0$, and choose a positive integer Q . For $\ell = 1, \dots, Q$, define

$$\omega_\ell = (2\ell - 1)\pi/T, \quad b_\ell = 4/[\pi(2\ell - 1)], \quad h_{\gamma,Q}(z) = \sum_{\ell=1}^Q b_\ell M_\gamma(\omega_\ell) \sin(\omega_\ell z).$$

Lemma 2 (Feature remainder). Put $d = \pi^2/(2\gamma T)$ and $u = d(2Q + 1)$. On $|z| \leq R_0$,

$$|\tanh(\gamma z) - h_{\gamma,Q}(z)| \leq e_{\gamma,Q}, \quad e_{\gamma,Q} := 4e^{-2\gamma(T-R_0)} + \frac{4\pi}{\gamma T} \frac{e^{-u}}{(1 - e^{-2u})(1 - e^{-2d})}. \quad (6)$$

Proof. The periodic square wave $\text{sign}(\sin(\pi z/T))$ agrees with $\text{sign}(z)$ on $(-T, T)$. Their difference has magnitude at most two outside. After smoothing, its magnitude is at most $1 - \tanh(\gamma(T - z)) + 1 - \tanh(\gamma(T + z))$, bounded by the first term of (6). The square wave has odd sine coefficients b_ℓ ; smoothing multiplies each by (5), giving an absolutely convergent series. For odd k , the smoothed coefficient equals $2\pi \text{csch}(dk)/(\gamma T)$. At $k \geq 2Q + 1$, use $\text{csch}(dk) \leq 2e^{-dk}/(1 - e^{-2u})$ and sum the geometric series with ratio e^{-2d} . This proves the second term. $\square$

Where the approximation enters. In (6), the first term pays for the artificial transitions at distance at least $T - R_0$ from our evaluation region. The second pays for the frequencies above the last retained harmonic. Even an infinite harmonic sum at fixed T represents a smoothed periodic square wave; increasing Q alone does not remove its distant-transition error. Choosing T, Q changes our mathematical representation, not the original network, center set, or training protocol.

Define three matrices, giving their entries so their roles are explicit:

- F has m rows and $2Q + 1$ columns: $1, \sin(\omega_1 x), \cos(\omega_1 x), \dots$, sampled at the inputs and divided by $\sqrt{m}$. It contains only the sample geometry.
- C has $2Q + 1$ rows and $W + 1$ columns. Its bias row is $(1, 0, \dots, 0)$, and its remaining bias-column entries are zero. For hidden column j , $C_{2\ell-1,j} = b_\ell \cos(\omega_\ell c_j)$ and $C_{2\ell,j} = -b_\ell \sin(\omega_\ell c_j)$. It contains only the center geometry and step coefficients.
- D_γ is diagonal with entries $1, M_\gamma(\omega_1), M_\gamma(\omega_1), \dots$. It contains all the gamma dependence.

The order of action is $\theta \mapsto C\theta \mapsto D_\gamma C\theta \mapsto F D_\gamma C\theta$: neuron coefficients become wave amplitudes, those amplitudes are attenuated, and the resulting function is evaluated at the inputs. F is an evaluation matrix, and C encodes center-dependent phases. Neither operation assumes an orthogonal projection into or out of Fourier space. The original readout weights remain the parameters; independently training all wave amplitudes would change the problem.

Proposition 2 (Finite factorization and kernel error).

$$\tilde{J}_\gamma = F D_\gamma C, \quad \tilde{K}_\gamma = F D_\gamma C C^T D_\gamma F^T. \quad (7)$$

With $\delta_\gamma = \sqrt{W} e_{\gamma,Q}$,

$$\begin{aligned} \|J_\gamma - \tilde{J}_\gamma\| &\leq \delta_\gamma, \\ \|K_\gamma - \tilde{K}_\gamma\| &\leq \Delta_\gamma := (2\|\tilde{J}_\gamma\| + \delta_\gamma)\delta_\gamma. \end{aligned} \quad (8)$$

Proof. Expand $\sin(\omega(x - c)) = \sin(\omega x) \cos(\omega c) - \cos(\omega x) \sin(\omega c)$ to obtain (7). The bias is exact, and each normalized hidden column has error norm at most $e_{\gamma, Q}$. Summing squared column errors bounds the Frobenius norm, hence the spectral norm, by δ_γ . Set $H = J_\gamma - \tilde{J}_\gamma$ and expand $K_\gamma - \tilde{K}_\gamma = H\tilde{J}_\gamma^T + \tilde{J}_\gamma H^T + HH^T$ to obtain (8). $\square$

The sampled waves in F need not be orthogonal, and CC^T need not be diagonal. Therefore $M_\gamma(\omega)^2$ is not generally an individual finite-kernel eigenvalue. Equation (7) retains both effects. It constructs the prediction using fixed geometry and explicit multipliers; it is not a fit to an optimization trajectory.

Exact operation after the approximation. Once $\tilde{J}_\gamma$ is defined, multiplying it by its transpose gives (7) exactly in real arithmetic. No off-diagonal entries of CC^T are dropped and no waves are declared to be kernel eigenvectors. The discrepancy from the original K_γ comes from approximating the features. The bound Δ_γ can overestimate that discrepancy because its proof uses Frobenius and triangle inequalities.

5 The missing implication: attenuation forces slow target energy

For any unit sample vector v ,

$$v^T K_\gamma v = \|J_\gamma^T v\|^2 = \sum_i \mu_i |u_i^T v|^2. \quad (9)$$

Thus small correlations with all features imply small spectral strength. The following theorem turns that observation into a target-dependent statement, including the effect of eigendirection rotation.

Choose a frequency cutoff $0 < \Omega \leq \omega_Q$. Let $P_{<\Omega}$ be the orthogonal projector, in the empirical Euclidean norm, onto the span of the bias and columns of F with frequency below Ω . Define $R_\Omega = I - P_{<\Omega}$ and

$$\alpha = \|R_\Omega y\| / \|y\|.$$

This is an explicit assumption on the target's component outside the sampled low-frequency span. It does not define the target using unknown slow eigenvectors. It also does not assert that a continuously high-frequency function is exactly orthogonal on a finite grid.

Let $F_{\geq\Omega}$ collect the remaining wave columns and $C_{\geq\Omega}$ the matching rows. Put

$$B_\Omega = \|F_{\geq\Omega}\|^2 \|C_{\geq\Omega}\|^2, \quad A_\gamma = \eta_\gamma [B_\Omega M_\gamma(\Omega)^2 + \Delta_\gamma].$$

Both the subspace and B_Ω are determined by fixed geometry, T, Q, Ω , independently of gamma and GD.

Theorem 1 (Gamma-dependent slow-target guarantee). For every $t \in (0, 1)$,

$$\begin{aligned} S_\gamma(t) &\geq (\alpha - \sqrt{A_\gamma/t})_+^2, \\ E_\gamma(n) &\geq (\alpha - \sqrt{A_\gamma/t})_+ (1-t)^n, \end{aligned} \quad (10)$$

where $(z)_+ = \max(z, 0)$. If $b_\gamma(t) := (\alpha - \sqrt{A_\gamma/t})_+ > \epsilon$, then

$$n_\epsilon(\gamma) \geq \left\lceil \frac{\log(b_\gamma(t)/\epsilon)}{-\log(1-t)} \right\rceil. \quad (11)$$

Proof. Step 1: gamma bounds action on the specified subspace. For a unit v in the range of R_Ω , all low-frequency coordinates of $F^T v$ vanish. Therefore

$$\begin{aligned} v^T \tilde{K}_\gamma v &= \|C_{\geq\Omega}^T D_{\gamma, \geq\Omega} F_{\geq\Omega}^T v\|^2 \\ &\leq B_\Omega M_\gamma(\Omega)^2. \end{aligned}$$

The diagonal multiplier decreases with absolute frequency. Applying (8) gives $v^T(\eta_\gamma K_\gamma)v \leq A_\gamma$.

Step 2: weak action forces the direction into slow modes. Let P_{fast} project onto rates $a_i > t$ and $P_{\text{slow}} = I - P_{\text{fast}}$. The spectral theorem gives $\eta_\gamma K_\gamma \succeq tP_{\text{fast}}$. Hence $\|P_{\text{fast}}v\| \leq \sqrt{A_\gamma/t}$.

Step 3: transfer that fact to the target. If $\alpha = 0$, (10) is trivial. Otherwise put $\hat{y} = y/\|y\|$ and $v = R_\Omega \hat{y}/\alpha$. Then $v^T \hat{y} = \alpha$. Orthogonality of the two spectral projectors and Cauchy–Schwarz give

$$\begin{aligned} \alpha &\leq |(P_{\text{slow}}v)^T P_{\text{slow}}\hat{y}| + |(P_{\text{fast}}v)^T P_{\text{fast}}\hat{y}| \\ &\leq \sqrt{S_\gamma(t)} + \sqrt{A_\gamma/t}. \end{aligned}$$

Rearranging proves the slow-mass bound. Corollary 1 proves the residual and time bounds. No shared eigenvectors across gamma, commuting operators, or identification of waves with eigenvectors was used. $\square$

Interpretation. Gamma enters the guarantee through the squared multiplier. The target enters through α , the step through η_γ , and the finite geometry through B_Ω and the approximation error. A positive statement requires $\alpha > \sqrt{A_\gamma/t} + \epsilon$. The mere presence of a tiny high-frequency component does not force a delay to a tolerance larger than that component.

A uniform common-slope cap corollary

For an explicit uniform statement, fix $0 < \gamma_- \leq \gamma \leq \bar{\gamma}$ and use the same geometry, T, Q, Ω , and rule $\eta_\gamma = \kappa/L_\gamma$ with $0 < \kappa \leq 1$. Let $V_{\bar{\gamma}, Q}$ be the second term of (6) evaluated at $\bar{\gamma}$, and define

$$\begin{aligned} e_* &= 4e^{-2\gamma_-(T-R_0)} + V_{\bar{\gamma}, Q}, & \delta_* &= \sqrt{W}e_*, \\ \Delta_* &= (2\sqrt{W} + 1 + \delta_*)\delta_*. \end{aligned}$$

Then Theorem 1 holds throughout this slope interval with

$$A_* := \kappa[B_\Omega M_{\bar{\gamma}}(\Omega)^2 + \Delta_*]$$

in place of A_γ . For any class of targets with $\alpha \geq \alpha_0$, replace α by α_0 to obtain a common learning-delay guarantee.

Proof. The distant-transition error is largest at γ_- . Every omitted Fourier coefficient is $b_\ell M_\gamma(\omega_\ell)$ with $b_\ell > 0$, so its absolute sum is at most the sum at $\bar{\gamma}$, bounded by $V_{\bar{\gamma}, Q}$. Thus the feature error is at most δ_* . Since $\|J_\gamma\| \leq \|J_\gamma\|_F \leq \sqrt{W} + 1$, expanding the kernel error around J_γ gives the displayed Δ_* . Finally $L_\gamma \geq 1$ implies $\eta_\gamma \leq \kappa$, and multiplier monotonicity gives $A_\gamma \leq A_*$ when the kernel-error bound is taken as Δ_* . Apply Theorem 1. $\square$

The positive lower slope endpoint makes this particular periodic approximation uniform. This corollary does not assert a sharp bound over all $0 < \gamma \leq \bar{\gamma}$, heterogeneous slopes, or varying center sets. Its constants can be large. We have not established that this new coarse corollary is tight on the reported experiment.

6 Where Theorem 1 loses information

The Fourier remainder and the slack in Theorem 1 are different quantities. The following steps identify the slack even if the approximate kernel were exact, so that $\Delta_\gamma = 0$.

Inequality 1: replace the actual high-frequency action by worst-case amplification. Write $F_H = F_{\geq \Omega}$ and similarly for C_H, D_H . The first proof step uses

$$\|C_H^T D_H F_H^T v\|^2 \leq \|C_H\|^2 \|D_H\|^2 \|F_H\|^2 \leq B_\Omega M_\gamma(\Omega)^2.$$

The first inequality allows the largest amplification of each matrix, even when the corresponding directions do not align. The second replaces every retained high-frequency attenuation by its largest allowed value. Both can overestimate the action on the particular direction v . Adding $\eta_\gamma \Delta_\gamma$ then pays for the original-to-Fourier kernel discrepancy, using a worst-case error bound.

Inequality 2: replace a distribution of fast rates by its cutoff. With $a_i = \eta_\gamma \mu_i$, the proof uses

$$t \|P_{\text{fast}} v\|^2 \leq \sum_{a_i > t} a_i |u_i^T v|^2 \leq v^T (\eta_\gamma K_\gamma) v \leq A_\gamma.$$

It thereby bounds $\|P_{\text{fast}} v\|$ by $\sqrt{A_\gamma/t}$ without retaining the individual rates or weights. Fast rates much greater than t can make this estimate loose.

Inequality 3: discard detailed target alignment. The proof uses Cauchy–Schwarz and bounds projection norms by one to obtain $\alpha \leq \sqrt{S_\gamma(t)} + \sqrt{A_\gamma/t}$. The actual angles between the target and the projected direction v can give a much larger slow mass than this guarantees. If $\alpha \leq \sqrt{A_\gamma/t}$, the resulting guarantee is zero, even when the actual target has slow energy.

Inequality 4: replace all slow-mode decays by one rate. Corollary 1 drops the fast-mode contributions to the residual and replaces each remaining $a_i \leq t$ by t . This gives $\sqrt{S_\gamma(t)}(1-t)^n$. Modes with $a_i \ll t$ actually persist much longer. This source of slack remains even when $S_\gamma(t)$ is computed exactly from a full spectrum. Figure 1 illustrates this last loss only; it does not evaluate inequalities 1–3.

These steps provide a general lower bound, not an approximate equality. Making the Fourier feature error small does not make their gaps small. We have not measured all of these gaps for the primary experiment, so the note makes no claim that any one is its dominant source of conservatism.

There is also no general matrix ordering from increasing the filter entries alone. For an algebraic example, take

$$G = \begin{pmatrix} 1 & 1 \\ 1 & 1 \end{pmatrix}, \quad D = \text{diag}(1, d), \quad 0 < d < 1.$$

Although the diagonal entries of D are at most those of the identity, $\det(G - DGD) = -(1-d)^2 < 0$. Thus $G - DGD$ is indefinite. For $v = (1, -1)/\sqrt{2}$, $v^T G v = 0$ but $v^T DGD v = (1-d)^2/2$. This is a counterexample to an inference from diagonal filtering alone, not a claim about the particular tanh geometry in the experiment. Target alignment and geometric coupling cannot be discarded.

Under additional simultaneous diagonalization, the simpler statement does hold: if an orthonormal frequency basis diagonalizes both the reference kernel and the smoothing operator, then the filtered kernel eigenvalues are $M_\gamma(\omega)^2 \mu_{\text{step}}(\omega)$ in that basis. Multiplying the two diagonal operators proves this immediately. Our finite model is not assumed to have this property. Equally spaced centers alone do not remove boundaries, sampling effects, or aliasing.

7 Compute the full spectrum of the Fourier-constructed kernel

This is the second branch, starting again from Proposition 2 rather than substituting Theorem 1’s lower bound. Fix T, Q and build F, C once. For each gamma, calculate D_γ and assemble $\tilde{J}_\gamma = F D_\gamma C$. Then compute its singular value decomposition,

$$\tilde{J}_\gamma = U \Sigma V^T, \quad \tilde{K}_\gamma = U \Sigma^2 U^T.$$

Thus the squared singular values give all nonzero kernel eigenvalues, and the columns of U give their sample-space eigenvectors. Project the target onto these vectors and retain its perpendicular residual. This is the full target-weighted spectrum. The plotted forecasts were calculated this way for every gamma. The implementation need not materialize the much larger sample-by-sample kernel to obtain its spectrum.

Here “full” means all modes of this finite Fourier-constructed matrix, with all center couplings and sample nonorthogonality included. The finite number of retained harmonics is still an approximation to tanh, controlled by Lemma 2. SVD and the following spectral sum are exact operations on that approximate model in real arithmetic; they do not use inequalities 1–4 from Section 6. Floating-point implementation error is discussed separately in Section 9.

Let $\tilde{K}_\gamma \tilde{u}_i = \tilde{\mu}_i \tilde{u}_i$ denote its positive modes, $\xi_i = \tilde{u}_i^T y$, and $y_\perp = y - \sum_i \xi_i \tilde{u}_i$. Set $q_i = 1 - \eta_\gamma \tilde{\mu}_i$ and assume both the true and approximate kernels satisfy the stable-step condition at the prescribed step. Define

$$\tilde{E}_\gamma(n)^2 = \frac{\|y_\perp\|^2 + \sum_i |\xi_i|^2 q_i^{2n}}{\|y\|^2}. \quad (12)$$

This is GD in the approximate filtered feature model. It is not a projection of the original GD residual onto a different basis. Proposition 1 proves its formula.

Which original-model information is used? The forecast uses the prescribed step, inputs, centers, target, and analytic multipliers. Its spectrum comes from $FD_\gamma C$. A separate SVD of the original tanh feature matrix supplies a reference comparison. The action refinement below also consults the original matrix to estimate the discrepancy; the analytic-remainder route does not require those directional actions. No decay rate is fitted to optimizer checkpoints. Numerically diagonalizing a new matrix for each gamma is part of this calculation, so this is a dictionary-specific prediction, not a closed scalar time law from the gamma cap alone.

Theorem 2 (Transfer of the full curve). Write $Z = K_\gamma - \tilde{K}_\gamma$ and let $\|Z\| \leq \Delta_\gamma$. Then $|E_\gamma(n) - \tilde{E}_\gamma(n)| \leq d_n$, where

$$d_n = \min \left\{ n\eta_\gamma \Delta_\gamma, \frac{\eta_\gamma}{\|y\|} \left[n\|Zy_\perp\| + \sum_i |\xi_i| \|Z\tilde{u}_i\| \sum_{j=0}^{n-1} q_i^j \right] \right\}. \quad (13)$$

Proof. Put $U = I - \eta_\gamma K_\gamma$ and $V = I - \eta_\gamma \tilde{K}_\gamma$. Both have norm at most one. The identity

$$U^n - V^n = \sum_{j=0}^{n-1} U^{n-1-j} (U - V) V^j$$

is obtained by expanding the sum and canceling adjacent terms. Since $U - V = -\eta_\gamma Z$, contraction bounds the norm of the sum by $\|U^n - V^n\| \leq n\eta_\gamma \Delta_\gamma$. Alternatively, apply the identity to y and substitute $V^j y = y_\perp + \sum_i \xi_i q_i^j \tilde{u}_i$. The triangle inequality and contraction of U give the second expression in (13). Finally, $\|U^n y - V^n y\| \leq \|(U^n - V^n)y\|$. Divide by $\|y\|$ and take the smaller bound. $\square$

The geometric sum equals $(1 - q_i^n)/(1 - q_i)$ for a positive rate, and its continuous limit is n at zero rate. The first bound uses the analytic feature remainder; the second also uses the original kernel’s action on known model directions. Neither needs optimizer iterates. Actions can be evaluated as $Zv = (J - \tilde{J})(\tilde{J}^T v) + J((J - \tilde{J})^T v)$ without subtracting nearly equal full kernel products.

Corollary 2 (Necessary and sufficient times). Put $\ell(n) = \max(0, \tilde{E}_\gamma(n) - d_n)$ and $u(n) = \min(1, \tilde{E}_\gamma(n) + d_n)$. If integers $n_- < n_+$ satisfy $\ell(n_-) > \epsilon$ and $u(n_+) \leq \epsilon$, then

$$n_- + 1 \leq n_\epsilon(\gamma) \leq n_+. \quad (14)$$

Proof. The lower test excludes the threshold at n_- and, by monotonicity, every earlier step. The upper test exhibits a step attaining it. The upper envelope need not itself be monotone; only the checked witness is used. $\square$

Bounds valid at different Q may be intersected. For positive finite intervals $L_g \leq n_\epsilon(g) \leq U_g$ and $L_h \leq n_\epsilon(h) \leq U_h$, division gives $L_g/U_h \leq n_\epsilon(g)/n_\epsilon(h) \leq U_g/L_h$.

Theorem 2’s transfer envelope can also have slack: its proof uses contractions and triangle inequalities, and the analytic Δ_γ may overestimate the actual discrepancy. The difference from Theorem 1 is that this slack surrounds a fully computed approximate-model curve. If that curve equals the true one and $Z = 0$, the transfer error is zero; Theorem 1’s inequalities can remain strict even in that situation.

8 The intermediate spectral evidence and the executed training

The primary experiment fixes 559 hidden features, 8,193 equally spaced training inputs on $[-1, 1]$, and equispaced centers with a halo. The target is

$$f^\star(x) = \sin(2\pi x) + \frac{1}{2} \sin(6\pi x) + \frac{1}{4} \sin(10\pi x).$$

All readouts start at zero. Gamma is 8, 12, 16, or 64, and the saved GD steps satisfy $\eta_\gamma L_\gamma \simeq 0.5$. The Fourier construction uses $T = 8$ and up to $Q = 2048$ odd harmonics. Its target weights come from a rectangular SVD of $FD_\gamma C$, with no fitted training rates. The plotted curves and masses below use $Q = 2048$; timing intervals can combine valid bounds from several Q values.

For clarity, write $\tilde{S}_\gamma(t)$ for the slow target mass of this approximate kernel. It is computed from its spectrum, including the perpendicular floor. Figure 1’s dashed bound is $\sqrt{\tilde{S}_\gamma(t)}(1-t)^n$ for the approximate residual $\tilde{E}_\gamma(n)$. It is Corollary 1 applied to the approximate model, not Theorem 1’s estimate $(\alpha - \sqrt{A_\gamma/t})_+(1-t)^n$. A lower bound for the original residual would additionally subtract the valid transfer error d_n and clip at zero; Figure 1 does not plot that transferred bound.

At rate cutoff $t = 10^{-6}$, the numerically computed slow target masses are 4.33333%, 0.00107533%, 0.0000285067%, and 0.000148493%, respectively. These are fractions of squared target norm expressed as percentages, not residual percentages. The cutoff was chosen as an explanatory diagnostic after the experiment, not as a precommitted selection criterion.

For gamma 8, Corollary 1 evaluated with the computed mass gives $\tilde{E}_8(16013) \gtrsim 0.20486$; the full-spectrum prediction is 0.22967. At gamma 64 the predicted residual is 0.00999980. For the approximate model, the same cutoff gives a gamma-8 necessary time of 3,035,752 updates, versus its full prediction and the executed tanh crossing of 15,798,313. This difference isolates the effect of using one rate cutoff on a spectrum we already computed in full. It is not a measured gap for Theorem 1. A single cutoff does not order all intermediate gammas: at this cutoff gamma 16 has less mass than gamma 64, despite taking longer to reach 1%.

The executed GD crossing counts are 15,798,313; 186,057; 61,792; and 16,013. The corresponding independently certified combined intervals are [15,784,048, 15,812,623], [186,057, 186,058], [61,792, 61,792], and [16,013, 16,013]. Their gamma-8/gamma-64 ratio is bounded by 985.70–987.49, versus a measured 986.59. All 1,796 saved GD residuals agree with the $Q = 2048$ filtered-model predictions within 8.92×10^{-15} in absolute relative residual. This numerical agreement is distinct from interval certification.

The analytic-remainder-only gamma-8 interval is [15,783,830, 15,812,843]; sharpness therefore does not require the action refinement. A control preserving the gamma-64 eigenvectors and relative eigenvalues, while rescaling only its largest eigenvalue to each case, predicts 16,013 updates at every gamma under the saved step rule. The observed separation requires deformation of the target-weighted normalized spectrum. All four runs attain 1%, so failure to represent that tolerance does not explain their different times.

The other archived targets are a single sine, an exponential of sine, a Runge function, and a quadratic. Across all five targets and four gammas, all 19 observed hits lie in their computed intervals; the quadratic/gamma-12 run stopped at 200,000 updates and is censored. Only the primary sine-mixture endpoints have the independent interval audit described next. These are retrospective training-optimization comparisons, not held-out generalization results.

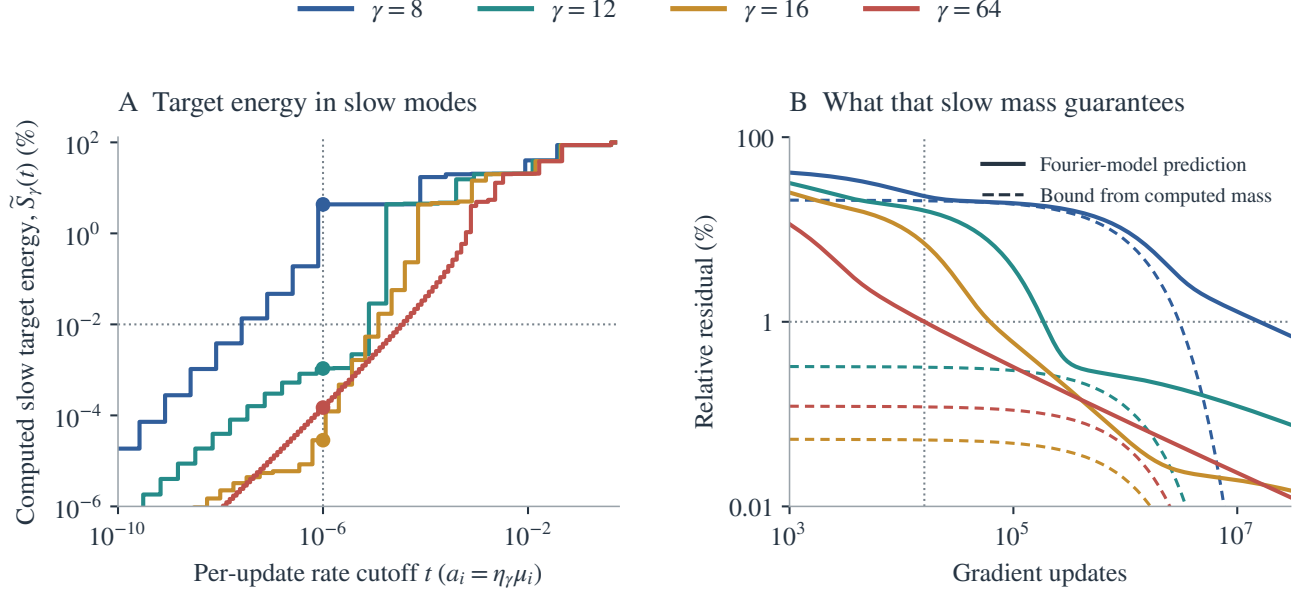


Figure 1: Computed spectral mass and the information lost by using one cutoff. A shows $\tilde{S}_\gamma(t)$ calculated from the $Q = 2048$ Fourier-constructed kernel; dots mark $t = 10^{-6}$. The horizontal guide is target energy 10^{-4} , or 0.01%, corresponding to a 1% residual. B compares the full spectral prediction $\tilde{E}_\gamma(n)$ with $\sqrt{\tilde{S}_\gamma(10^{-6})}(1 - 10^{-6})^n$. This dashed bound uses the computed mass, not Theorem 1’s analytic estimate. It bounds the approximate model; the true-kernel transfer error is not subtracted here. The horizontal guide is a 1% residual and the vertical guide is 16,013 updates. All curves are FP64 kernel calculations, not executed trajectories or independently interval-certified spectral masses.

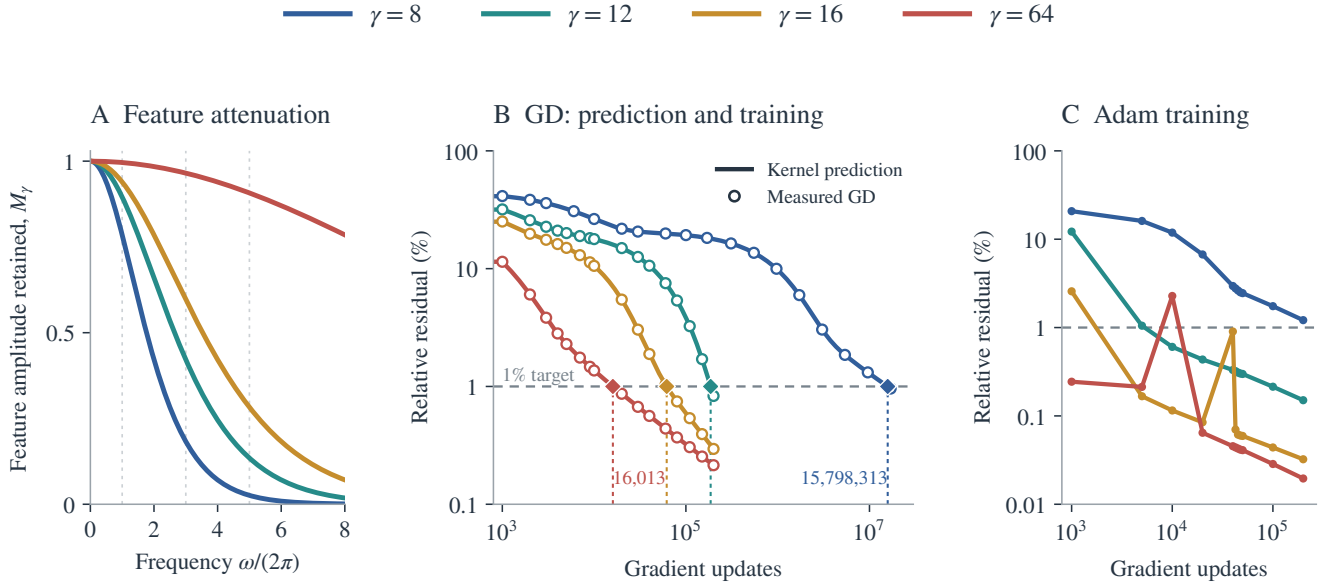


Figure 2: Full spectral forecasts compared with executed optimization. A evaluates the analytic multiplier; frequency guides mark the three target frequencies. B’s lines are the complete spectral GD sum for $FD_\gamma C$ at $Q = 2048$, not Theorem 1’s lower bound or an error envelope. Open circles are actual saved GD residuals; diamonds mark measured 1% crossings. Dotted verticals locate approximate-model crossings. The certified intervals reported in the text additionally account for discrepancy from the tanh model. C connects actual Adam checkpoints under the same optimizer settings across gammas; the GD proof does not apply to Adam. Both optimizers start at zero. GD measurements are subsampled by update count for visibility, with every saved residual retained in the evidence.

9 What exactly was certified numerically?

The analytic proofs above are exact-arithmetic statements. FP64 feature assembly, SVD reconstruction, and action estimates require additional numerical error control. The archived calculation records sensitivity allowances, but these are not automatically interval enclosures of every plotted quantity. In particular, Figure 1’s masses and cutoff bounds have not received an independent interval certificate.

The primary timing endpoints were separately verified using 192-bit Arb ball arithmetic on nominal real tanh, with the archived binary inputs, centers, target values, and step interpreted exactly. The independent checker uses neither the SVD nor executed GD trajectories. Its algebra is as follows.

Put $H = J_\gamma^T J_\gamma$ and $g = J_\gamma^T y$. The coefficient recurrence is $\theta_{n+1} = (I - \eta_\gamma H)\theta_n + \eta_\gamma g$. Hence, defining

$$\mathcal{B} = \begin{pmatrix} I - \eta_\gamma H & \eta_\gamma g \\ 0 & 1 \end{pmatrix}, \quad \begin{pmatrix} \theta_n \\ 1 \end{pmatrix} = \mathcal{B}^n \begin{pmatrix} 0 \\ 1 \end{pmatrix},$$

gives the exact identity

$$E_\gamma(n)^2 = \frac{\|y\|^2 - 2\theta_n^T g + \theta_n^T H \theta_n}{\|y\|^2}. \quad (15)$$

Both follow by induction and expansion of $\|y - J_\gamma \theta_n\|^2$. Integer matrix powers evaluate (15) without stepping through an optimizer trajectory.

To construct the common-slope Gram matrix efficiently, let m_j be the empirical mean of $\tanh(\gamma(x - c_j))$. For distinct centers,

$$H_{jk} = 1 - \frac{m_j - m_k}{\tanh(\gamma(c_k - c_j))}.$$

Indeed, the tanh subtraction identity gives $a - b = \tanh(\gamma(c_k - c_j))(1 - ab)$ for the two features at any input; average and rearrange. Diagonal entries and repeated-center pairs use direct products; $H_{00} = 1$ and $H_{0j} = m_j$.

The audit first encloses $\eta_\gamma \|H\|$ below one using a row-sum bound. For a proposed interval $[N_-, N_+]$, it then encloses (15) at the excluded step $N_- - 1$ and sufficient step N_+ . An enclosure strictly above the exact threshold $1/10000$ excludes the first, while an enclosure at or below it confirms the second. Proposition 1’s monotonicity completes the certificate. The reported certificates concern these endpoints, not every FP64 curve, target, or newly stated coarse cap bound. They also distinguish the nominal-real problem from floating-point executed GD, whose measured crossings are compared separately.

10 Bandwidth scaling and the final scope

When centers have spacing h , keep $\lambda = \gamma h$ for dimensionless bandwidth. At a frequency fixed relative to the grid, $\vartheta = \omega h$,

$$M_\gamma(\vartheta/h) = \frac{\pi|\vartheta|/(2\lambda)}{\sinh(\pi|\vartheta|/(2\lambda))}.$$

Substitution into (5) proves that fixed λ preserves attenuation at a fixed grid-relative frequency. For $h = 2/N$ and $W = N + 2\lceil\sqrt{N}\rceil + 1$, fixed positive λ implies $\gamma = \lambda N/2 = \Theta(W)$. This does not imply constant training time across widths: the target’s sampled spectrum, geometry, and normalized rates still enter (1).

The shared starting point is gamma’s explicit feature multiplier. Theorem 1 converts it to a lower bound through geometry norms, a target-overlap condition, and four sources of inequality slack. The plotted forecast instead constructs a finite Fourier feature matrix for each gamma, computes its SVD, keeps all target weights, and uses Theorem 2 to transfer its curve to the tanh model. Close agreement is expected when that matrix accurately approximates the original one; it validates the quantitative construction and timing calculation, but does not establish sharpness of Theorem 1 or a universal scalar law for a gamma cap. The numerical evaluation of that coarse bound on the primary geometry remains separate, unfinished work.

Sources and reproduction

The proofs are self-contained apart from standard finite-dimensional spectral theory, Fourier series, and the residue theorem. The classical GD spectral interpretation is consistent with Y. Yao, L. Rosasco, and A. Caponnetto, [On Early Stopping in Gradient Descent Learning](#), *Constructive Approximation* 26:289–315 (2007), §3.3. The contribution being evaluated is the explicit gamma-to-finite-kernel connection and its target-dependent quantitative consequences.

The [study report](#), [interval audit](#), [spectral figure data](#), and [executed-curve provenance](#) retain the inputs and evidence roles. Earlier detailed constructions remain in [the technical note](#); its historical eigenvalue notation differs from this note.

Reproduce Figure 1 with the [spectral figure script](#) and Figure 2 with the [GD/Adam figure script](#). Compile the directly authored [LaTeX source](#) with `latexmk -pdf` from the repository root. The [Markdown companion](#) contains the full argument and commands. This write-up and spectral diagnostic use the existing archive and no additional GPU training.